Participatory Analytics

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AGILE – PhD School
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Engineer by training

Geodesy and Cartography

PhD in Geoinformatics

Participatory Analytics, participatory mapping, and geospatial data management

Agenda

Hands-on Workshop:

1- Intro to a participatory mapping guide for ethnic groups

2- Intro to Sketchmap Tool and its basemap options

3- Identify the study area to map urban deprivation based on morphology

- Pereira, Colombia.

4- Comparison between mapped areas and IDEAMAPS models

5- Contrast between mapped areas and earthquake impacts reported by communities

6- Reflection, emergent approaches and wrap up

Mapping Methodology for T-622 Action Plan

Guidelines for Project Planning Workshops

Context to Ruling T-622

- The Colombian Constitutional Court issued Ruling T-622 in 2016. The Ruling recognised the legal personhood of the Atrato River and demanded the restoration, protection and conservation of the river.
- T-622 established the Guardians of the Atrato to represent communities and the Atrato River in the process for implementing the ruling.
- The action plan for the environmental recovery of the basin was approved by the communities and, after a period of diagnostics and institutional preparation, its implementation has already started. However, the specific actions of the plan require ongoing consultation with the communities.
- Now, projects are being proposed in support of the action plan. It is important that communities are central to the planning and implementation of these projects.

3

Why consult communities?

- **Legal requirement**: Afro and Indigenous communities must be consulted on projects implemented on their collective territories.
- **Local knowledge**: communities possess a wealth of knowledge about their collective territories. Incorporating such knowledge into the policy process is pivotal to developing effective projects that are supported by communities.
- **Accountability**: it is important that community representatives have knowledge and understanding of projects within their territories to hold policy actors to account.

4

Why map?

- The Mapping Methodology facilitates dialogue between communities, their allies, and policy actors, such as IIAP and Minambiente, during the planning of projects in support of the T-622 action plan.
- Representatives and associates of the Cuerpo Colegiado de Guardianes lead workshops to ensure the voices of communities are heard.
- Communities, their allies and policy actors, such as IIAP, use simple mapping tools during the workshops to exchange knowledge around the planning for projects.
- By the end of each workshop, the participants agree on a set of action points for what happens next with the proposed project.

Roles and responsibilities

- When a policy actor, such as IIAP, has identified a potential location for a new project, a workshop should be organised with communities of that territory.
- Workshops are led by representatives and associates of the *Cuerpo Colegiado de Guardianes*. **They assume the role of workshop facilitators.**
- To ensure meaningful discussion, the maximum number of participants for a workshop should be around 30 people.
  - Members of the community: participants should reflect the diversity of the community, along lines of gender and age, for example.
  - Allied organisations: representatives of accompanying organisations may be invited to support community participation.
  - Policy actors: representatives of the relevant policy organisations must be present to exchange knowledge with the community.
- It is recommended that the workshop is held over the course of one-day (approx. seven hours in total).
- **The safety of participants is paramount.** To ensure that risks are minimised, there must be careful consideration of the location of workshops, the invitation list, the confidentiality of participants and sharing of outputs. The Cuerpo Colegiado de Guardianes and relevant policy actors are responsible for discussing these issues, and putting in place measures to minimise risk.

1. Introduction to the Workshop (30 mins)

- All participants in the workshop briefly introduce themselves.
- The facilitator explains that participation in the workshop is voluntary and outlines how information from the workshop will be used.
- The facilitator then provides context for the workshop and outlines the agenda for the day. It is important to emphasise the following points:
  - Communities hold extensive knowledge about their territories. This knowledge should be valued and integrated into projects.
  - The workshop provides a platform for policy actors to explain and discuss their plans with local communities.
  - Maps are an important way of communicating information. Through dialogue and mapping, participants can share knowledge.
  - The workshop is not just about discussion, though. By the end of the workshop, participants will agree on action points for next steps.

9

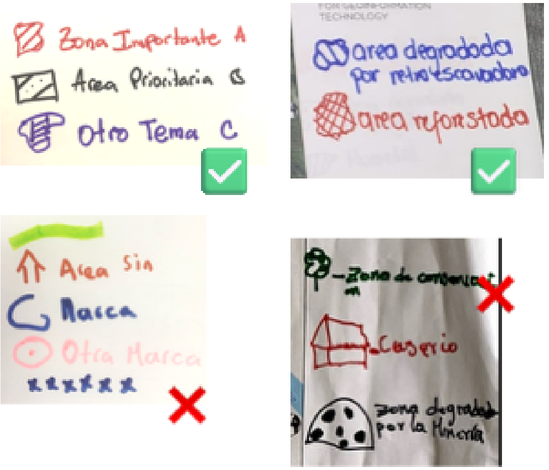
2. Discuss the aim of the workshop (45 mins)

- Objective: reach consensus on the aim for the workshop.
- These guidelines are designed to be used when a policy actor has funding for a specific purpose. The aim of the workshop will thus closely align with this purpose.
  - Example: IIAP receives funding to implement reforestation projects in Rio Quito. The workshop participants decide the aim of the workshop should be to identify priority areas for reforestation.
- This section of the workshop opens with representatives of the policy actor explaining their goals to the other participants, e.g. outline purpose of funding, levels of resources and timeline for planning and implementation. The policy actor then proposes an aim for the workshop.
- Discussion: the facilitator invites responses and questions from the room. Participants may suggest changes to the proposed aim and/or offer alternative aims.
- The facilitator finds consensus in the room, sets the aim of the workshop and writes it on the flipchart.

10

3. Identify evidence needs (45 mins)

- Objective: agree on three types of information to be captured during mapping.
- Participants now need to discuss what types of information are needed to achieve the aim of the workshop. The participants will mark this information on the maps during the afternoon session.
  - Example: to identify priority areas for reforestation in Rio Quito, the workshops participants agree that it is important to know the location of agricultural lands, sites of biocultural significance and abandoned mining areas.
- The facilitator divides the participants into smaller groups of approx. five people. The groups should include a mix from the different sectors represented in the room. One person should be assigned the role of group lead, taking notes on the discussion to report back. Prompt questions:
  - What type of information would be needed to achieve the aim of the workshop?
  - What sort of information would the policy actor typically collect for this kind of project?
  - What knowledge do the community posses that would be important to the aim of the workshop?
- Each group reports back the main points of their discussion. Participants are given time to ask questions.
- The facilitator finds consensus in the room and notes a maximum of three types of information that will be captured on the maps. These types of information will form the basis of a map legend – see next page.



Creating a map legend

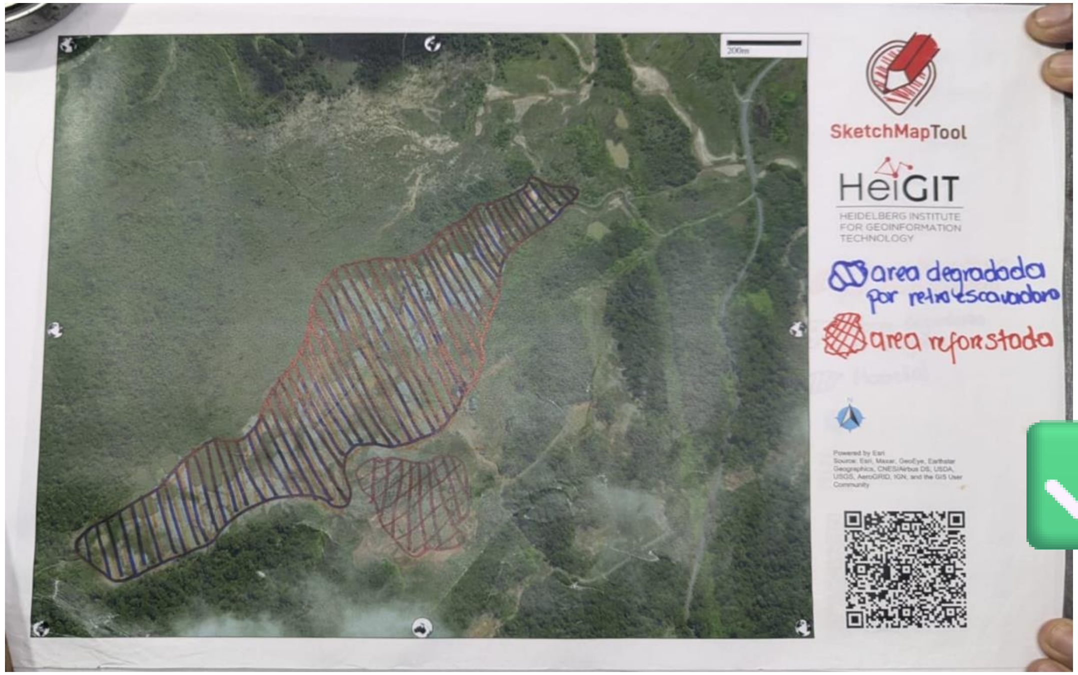
- A map legend describes the types of information that will be captured on the maps. The groups will use the legend to draw on the maps.
- Assign a colour to each type of information from stage 3: red, blue and black.
- On a flipchart, the facilitator draws symbols for each of the types of information, using a different colour for each.
- These symbols should be simple and filled-in with lines, as shown in the top two examples. This is important to ensure that the SketchMap Tool can recognise the drawings when the maps are uploaded during the final stage.
- As shown in the bottom examples, intricate designs and/or use of colours that do not contrast well with the printed maps should be avoided.
- The mapping legend created by the facilitator can then be used by all groups in the next stages.

CUERPO COLEGIADO DE GUARDIANES DEL ATRATO
FORO INTERÉTNICO SOLIDARIDAD CHOCÓ
SCIAF Uniting for a just world Caritas Scotland
SIEMBRA Centro Sociojurídico para la Defensa Territorial
Urban Big Data Centre
University of Glasgow College of Social Sciences

Introducing SketchMap Tool

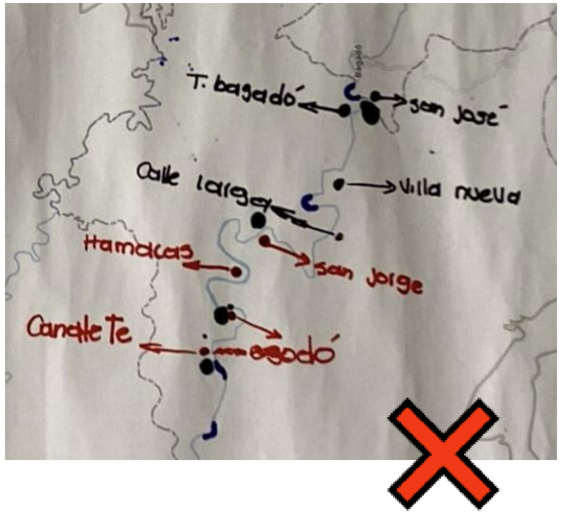
An easy-to-use system for participatory mapping

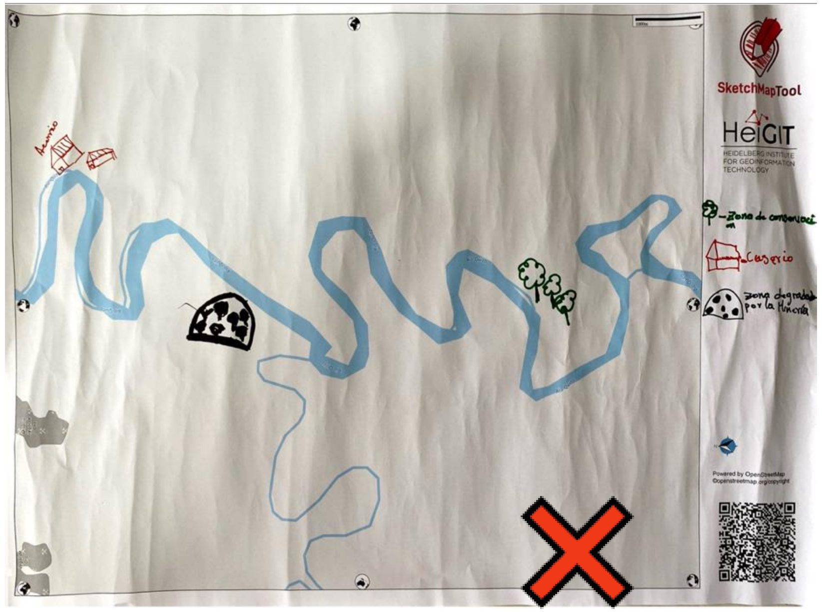
- SketchMap Tool is a collaboration between the University of Glasgow (UBDC in Glasgow) and the Heidelberg Institute for Geoinformation technologies (HeiGIT).
- It is an open-source web application that allows users to create their own maps.
- Overview of the process:
  - Users generate a map on the SketchMap Tool website and print it.
  - Users add to the map by drawing on the physical copy.
  - Finally, they take a picture of the printed map and upload it to the SketchMap Tool website. This step creates a digital version of the map.
- Full instructions on how to use SketchMap Tool as part of the Mapping Methodology follow this page.



Drawing on the Map

- Groups draw the map legend from step 3 in the margin of the map, as shown in the first two images.
- Before drawing on the map, groups discuss the types of information to be added to the map, e.g. Where are the areas of deforestation? When did the deforestation occur?
- The groups then start to draw on the maps using the map legend, as shown in these two images.





Tips

To ensure that the SketchMap Tool can recognise the drawings when the maps are uploaded during the final stage:

- Follow the map legend and use the three approved colours: blue, black and red.
- Use thick lines and close the areas.
- You can overlap drawn areas, if using different colours, but be sure to close the areas.
- Avoid the addition of writing, icons or complex drawings to the map, as these won't be picked up by the application when the maps are uploaded.
- Some of these common errors are shown here.

Sketchmap, teamwork:

1- Get familiar with the area in the printed maps

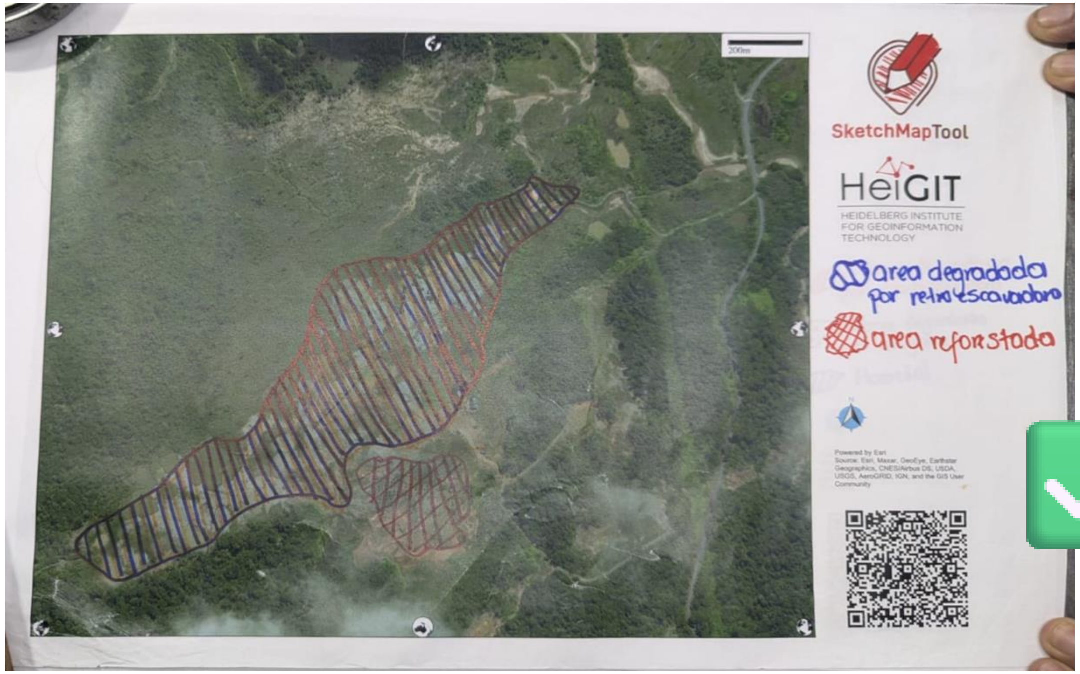
2- Explore the definition of “morphological informality”

Hint: www.ideamapsdataecosystem.org

3- Discuss impacts of the latest earthquake hitting western Colombia

Hint: Pereira, Colombia.

Hint: Chocó, Colombia



1

- Using a camera phone, take photos of all of the maps.
- All of the photos must include the QR code and the special marks that surround the maps.
- The photos should be taken in good light, but avoid glare, as this will make it difficult for the application to recognise the drawings.

2

- Open SketchMap Tool again: https://sketch-map-tool.heigit.org
- On the landing page of the website, select 'Digitaliza tus Sketch Maps'.

3

- On this page, the photos may be uploaded to SketchMap Tool.
- The uploading process may take a few moments.
- Once it is complete, the application will give the digital file as GeoTiff and GeoJSON. Both can be used in GIS applications for different purposes.

4

- To protect the confidentiality of the workshop, we recommend refusing permission for SketchMap Tool to keep copies of the maps.
- Once the files have been downloaded, save them to your device and transfer them to the Cuerpo de los Guardianes.

Digitising Sketchmaps, teamwork:

1- Take a photo of the map and your sketches

2- Process the photo (jpg / png format) via SketchMapTool.org

3- Discuss about the results.

    Hints: data formats, completeness, accuracy

4- Discuss how this would fit in a research setup

    Hint: Tech requirements

    Hint: Data/map literacy

    Hint: Use cases

Compare sketched areas and IDEAMAPS

1- Explore the Zenodo Community

2- Download the modelled dataset

Hint: https://zenodo.org/records/17886610

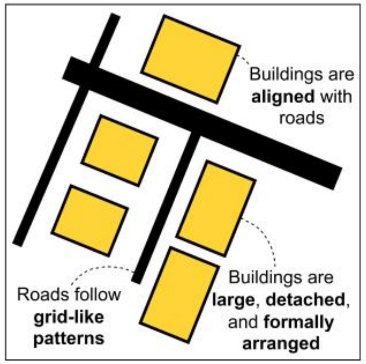
3- Discuss how the areas and their representations differ

IDEAMAP's lay definition

morphological informality: represents the difference between planned and unplanned areas of the city as indication of urban deprivation. Areas with higher levels of morphological informality are related to more deprived areas

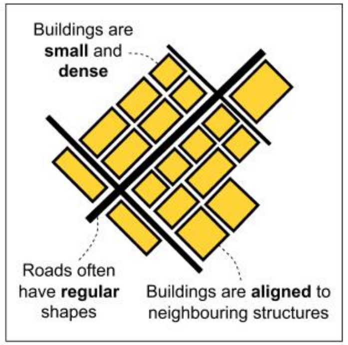
Low

Buildings are larger, detached from one another, and regularly arranged. It is easy to visualise roads, pathways, or open spaces between them. Areas with no buildings are also categorised as low.

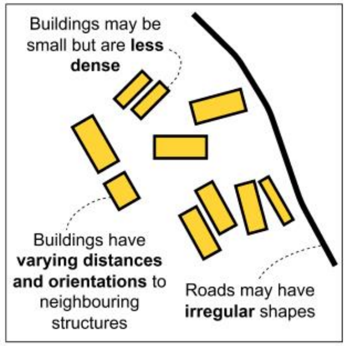


Medium

Buildings are small and close together (it can be hard to visualise roads, pathways, or spaces between them) OR buildings are irregularly arranged (with varying alignment to neighbouring structures). Roads are moderately present.

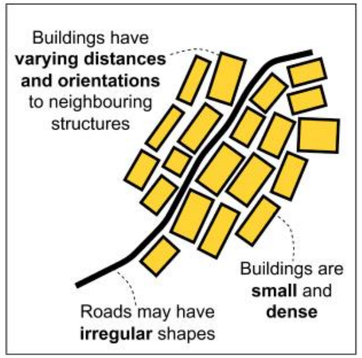


OR



High

Buildings are small and close together AND buildings are irregularly arranged (with varying alignment to neighbouring structures). Roads are hard to visualise or they are absent.



How our perceptions differ from existing models?

zenodo

Search records... Communities My dashboard diegopaja...

Urban Big Data Centre

Published December 10, 2025 | Version v1

Dataset Open

Morphological Informality Modelled Dataset in Pereira, Colombia

Hafner, Sebastian (Researcher)[1]; Zhao, Qunshan (Researcher)[2];
Abascal, Angela (Researcher)[2]; Comerio de Paulo, Manuella (Researcher)[2];
Tregonning, Grant (Researcher)[2]; Middleton, Alexandra (Researcher)[3];
Shonowo, Adenike (Researcher)[2]; Kuffer, Monika (Researcher)[4];
Engstrom, Ryan (Researcher)[5]; Thomson, Dana R (Researcher)[6]; *et al.* Show all 18 authors

Contributors

Data curator: Pajarito Grajales, Diego Fabian[1]

Show affiliations

This dataset represents the difference between planned and unplanned areas of the city as an indication of urban deprivation. Areas with higher levels of morphological informality are related to more deprived areas.

Produced as part of the IDEAMAPS Data Ecosystem project, the work was supported, in whole or in part, by the Bill & Melinda Gates Foundation INV-045252. Under the grant conditions of the Foundation, a Creative Commons Attribution 4.0 Generic License has already been assigned to the Author Accepted Manuscript version that might arise from this submission. The findings and conclusions contained within are those of the authors and do not necessarily reflect positions or policies of the Bill & Melinda Gates Foundation.

Files

Edit

New version

Share

Versions

Version v1 Dec 10, 2025

10.5281/zenodo.17886610

Cite all versions? You can cite all versions by using the DOI 10.5281/zenodo.17886609. This DOI represents all versions, and will always resolve to the latest one. Read more.

University of Glasgow

Urban Big Data Centre

Let’s do some “science”

Count how many grid cells fall into the area you sketched.

Hint: intersect

Visualise the proportion of cells classified in Low / Medium / High

Hint: Bar chart

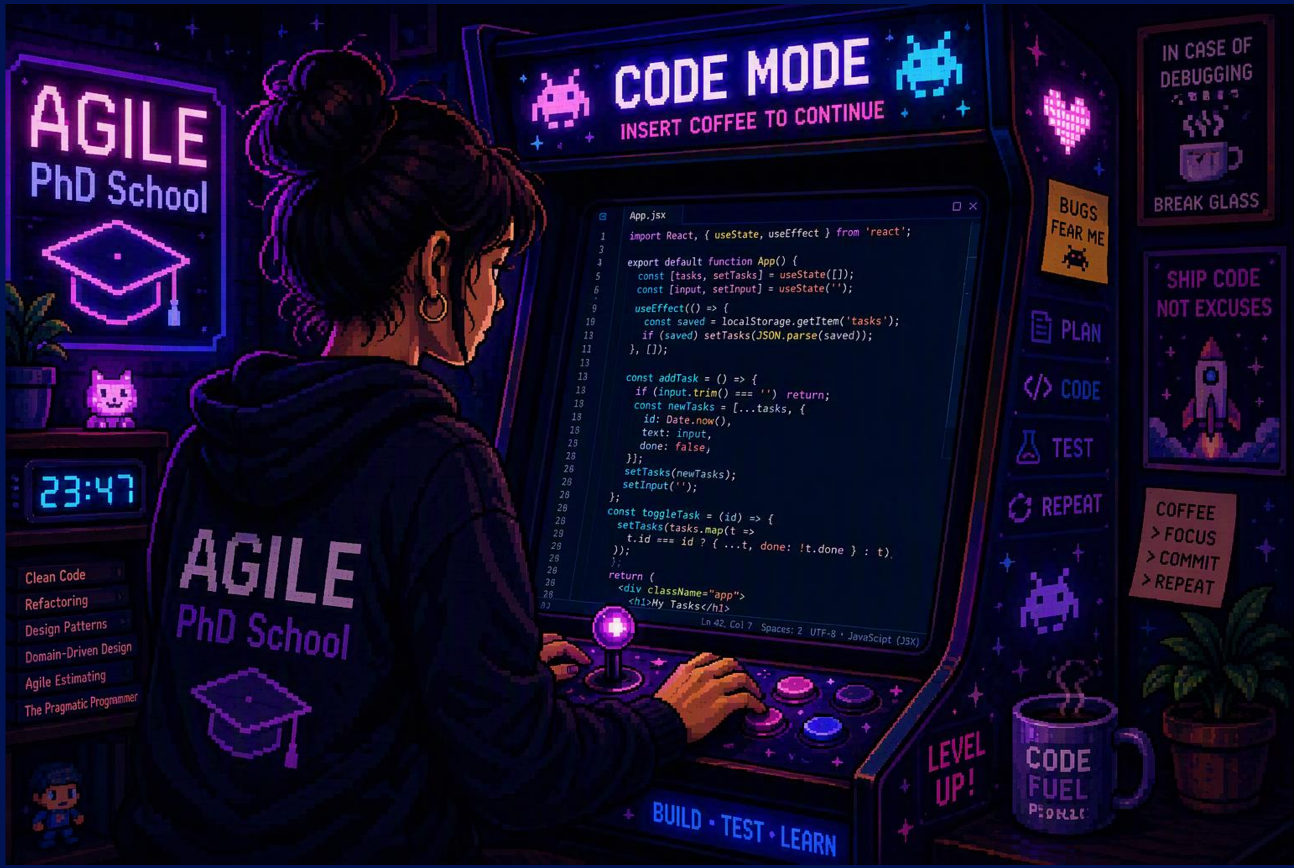


Image generated with Microsoft Copilot using AI-assisted image creation.

How our perceptions differ from existing models?

Image generated with Microsoft Copilot using AI-assisted image creation.

Compare against a third dataset

1- Is there any relationship between urban informality and disaster risk management

2- Get familiar with the recent earthquake hitting Colombia

3- Navigate citizen-generated data about impacts of such an earthquake

4- Build some numbers or maps that support your findings

5- Share your 1-slide / image with the group

How our perceptions differ from existing models?

Abstract

The relationship between informality and vulnerability, common in Latin American communities, is analyzed using data from two medium-sized cities in the Chilean urban system as case studies: Iquique, located in the northern desert zone of the country, and Puerto Montt, located in the rainy south. Informal settlements, or '*campamento*' (literally meaning campsites in Chile), within these cities were studied and analyzed. The results revealed that while these originally illegal *campamentos* have gradually been absorbed by urban sprawl and have reached internal consolidation, they continue to maintain their conditions of precariousness and social vulnerability.

The *campamento* settlements are frequently located in areas exposed to natural hazards. They have also been subject to institutional backwardness, or a lack of oversight, with little to no attention being paid to risk reduction. In Iquique, the *campamentos* located in the northern zone of the city are highly exposed to seismic threats and landslides. They are also close to the port and the tax free zone (ZOFRI). In the case of Puerto Montt, the most vulnerable areas of the *campamentos* are the coastline areas to the west and east of the city, which are subject to landslides and flooding.

The *campamento* settlements constitute real vulnerability *hotspots* for their residents, as they are high risk zones within the cities, characterized by high levels of precariousness. In general, they have not been considered in the urban improvement measures, even though they have been part of each city's main urban area for several decades now.

ELSEVIER

International Journal of Disaster Risk Reduction

Volume 13, September 2015, Pages 109-127

Disaster risk construction in the progressive consolidation of informal settlements: Iquique and Puerto Montt (Chile) case studies

Carmen Paz Castro [a], Ignacio Ibarra [a], Michael Lukas [a], Jorge Ortiz [a], Juan Pablo Sarmiento [b]

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https://doi.org/10.1016/j.ijdrr.2015.05.001

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Paz Castro et al., 2015
https://doi.org/10.1016/j.ijdrr.2015.05.001

At least 169 killed in Colombia's largest earthquake in years

In shattered Cali, residents search for loved ones

José Carlos Cueto , BBC News Mundo correspondent, Bogotá, **Vanessa Buschschlüter**, Latin America online editor and **Michael Sheils McNamee**

10 August 2026

Updated 11 August 2026

Colombia: Earthquake Causes Significant Damage in Pereira

Pereira, COlombia, Aug. 10, 2026. X/ @Updatesofwworld

August 10, 2026 Hour: 10:42 am

Six Airports Report Damage as Authorities Assess Destruction and Injuries Across Several Cities

The magnitude 7.4 earthquake that shook much of Colombia on Monday caused significant material damage in Pereira, capital of the department of Risaralda. The earthquake, which occurred at 7:34 a.m. local time, was centered in the town of San Jose del Palmar, in Choco, at a depth of 82 kilometers, the Colombian Geological Service (SGC) said.

Temblor Colombia 🇨🇴 | MY MAPS

All

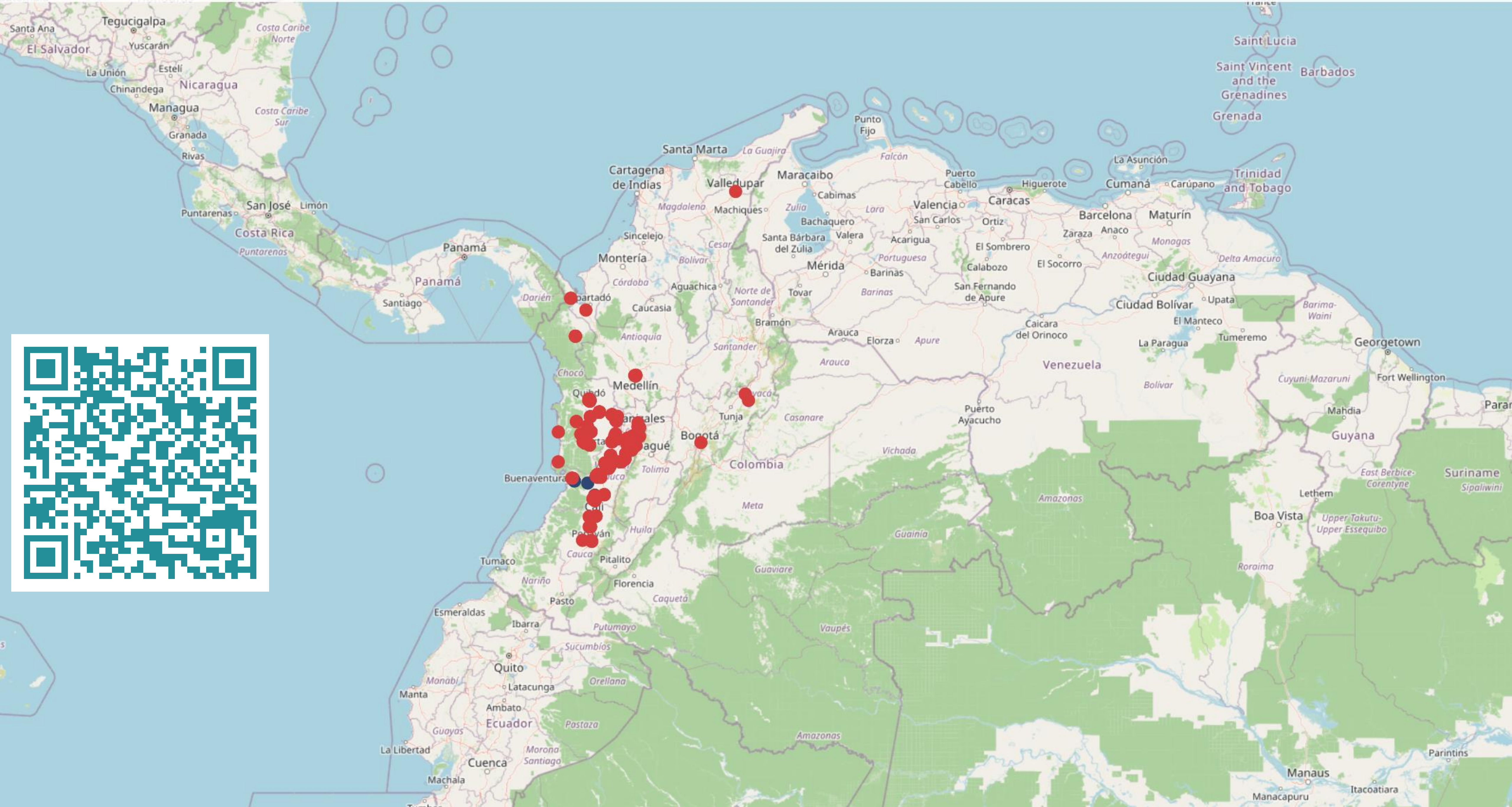

Image generated with Microsoft Copilot using AI-assisted image creation.

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Diego Pajarito Grajales

Co-production Session at 13:30

AI-assisted image creation.

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September 2026

TRANSFERRING

DATA2KNOWLEDGE

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Diego Pajarito Grajales | Urban Big Data Centre, University of Glasgow, UK

Co-designed with: Ana Maria Bustamante Duarte | Faculty of Architecture and Design, Universidad de Los Andes, Colombia

MAPPING PLACES

ORIGIN AND RESEARCH

STRENGTHENING THE

INNOVATION FACTOR IN YOUR RESEARCH

AGILE PHD SCHOOL

GOAL FOR DAY 3 11 AM

To deliver a presentation for a **joint Research Proposal** that builds upon the individual **PhD interests and methods**

Research Question
Roles within the team
Stakeholders
Timeline
Budget

REFINING YOUR RESEARCH QUESTION FOR INNOVATION

THEN, WHEN CHOOSING WE HAVE TO THINK ABOUT IT BEING...

- **Focused** (narrow enough to be manageable).
  - It has to have a clear spatial and thematic focused.
- **Relevant/Motivating**(for/from society and us).
  - Address real-world problems and/or fills significant knowledge gaps.
- **Researchable** (it can be investigated with available methods and data).
- **Original** (it offers a new perspective or fills a gap).
- **Grounded** (it is supported by existing literature and theory).
- **Feasable** (context, technology/technical abilities, time constraints, funding).
- **With innovation potential**
  - This eases continous work, funding, collaboration, and it helps with commercial viability if needed.

SUGGESTED STRUCTURE OF RESEARCH QUESTIONS

Research questions should be **formulated to encourage in-depth exploration and critical thinking**. If a question can be answered with "yes" or "no," it is not correct. Instead, **use** open-ended question words such as **how, why, which, or what** to prompt more meaningful responses.

Product innovation

- New or improved goods/services.

Social innovation

- New solutions to social problems.

Technological innovation

- Novel technical solutions.

Process innovation

- New methods or approaches.

Business model innovation

- New ways of creating/delivering value.

SUGGESTED STRUCTURE OF RESEARCH QUESTIONS

INTRO

Specify the problem or innovation gap you are addressing (Example: Given that... Due to....)

QUESTION

how might we develop *(type of innovation)* through *(methodological approach)* OR that enables *(specific capabilities for the development)* + for *(target users)* + in/at *(context)*

MAPPING

Region City

Country

- Africa
- Antarctica
- Asia
- Europe
- The Americas
- Australia/Oceania

Indonesia

Bali

S
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M
E

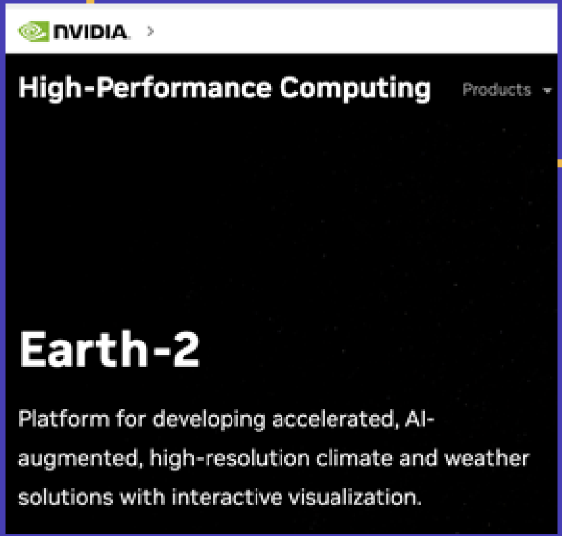
TIPS

INTRODUCING

R4I

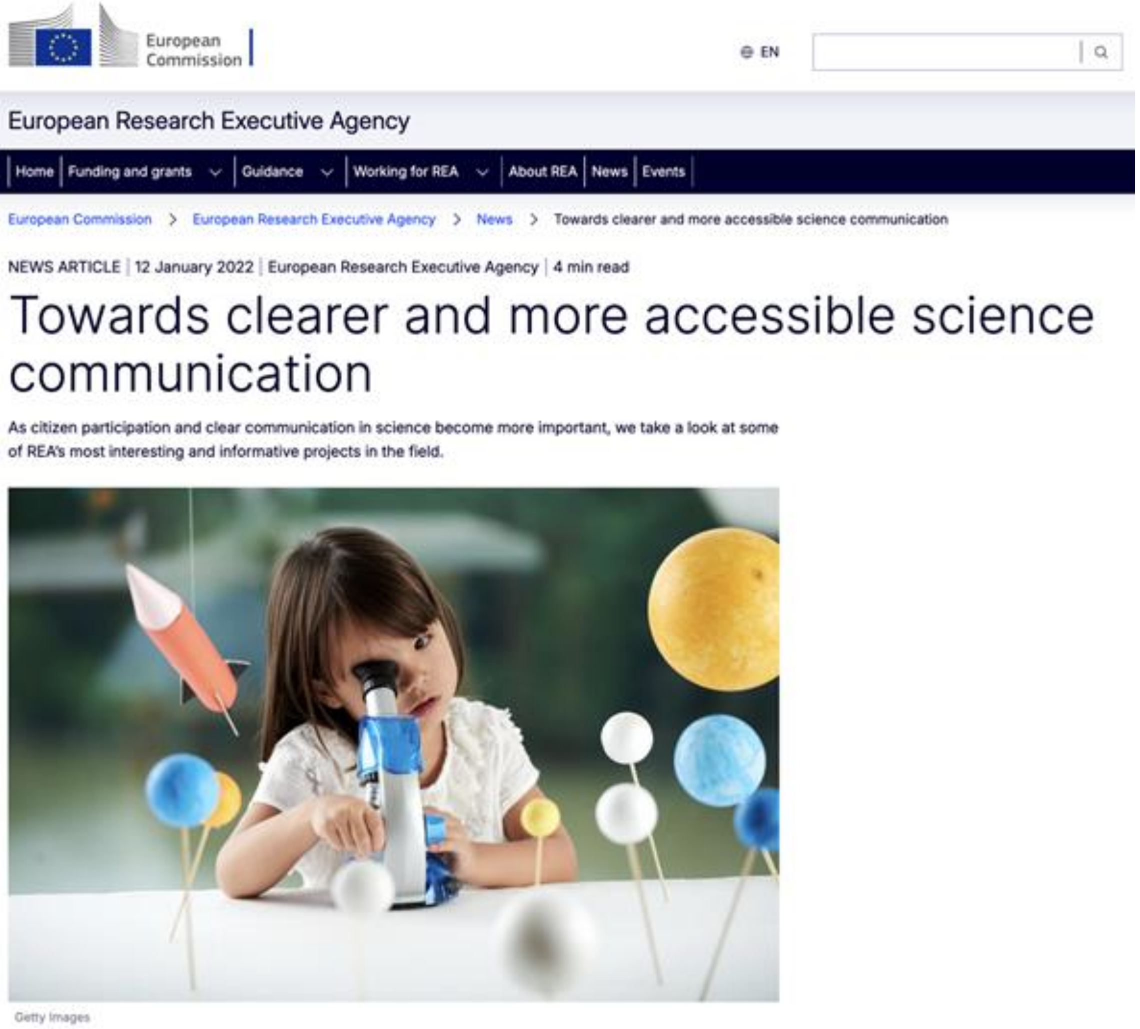
YOUR OPENING ....

What if an AI Foundation Model could Simulate Global Climate at Kilometer-Scale Resolution?



"We acknowledge the protagonism of the volunteers and communities who took up our approach with enthusiasm. We're delighted to receive this recognition"

Use language that excites both technical and non-technical stakeholders


European Commission

EN

European Research Executive Agency

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European Commission > European Research Executive Agency > News > Towards clearer and more accessible science communication

NEWS ARTICLE | 12 January 2022 | European Research Executive Agency | 4 min read

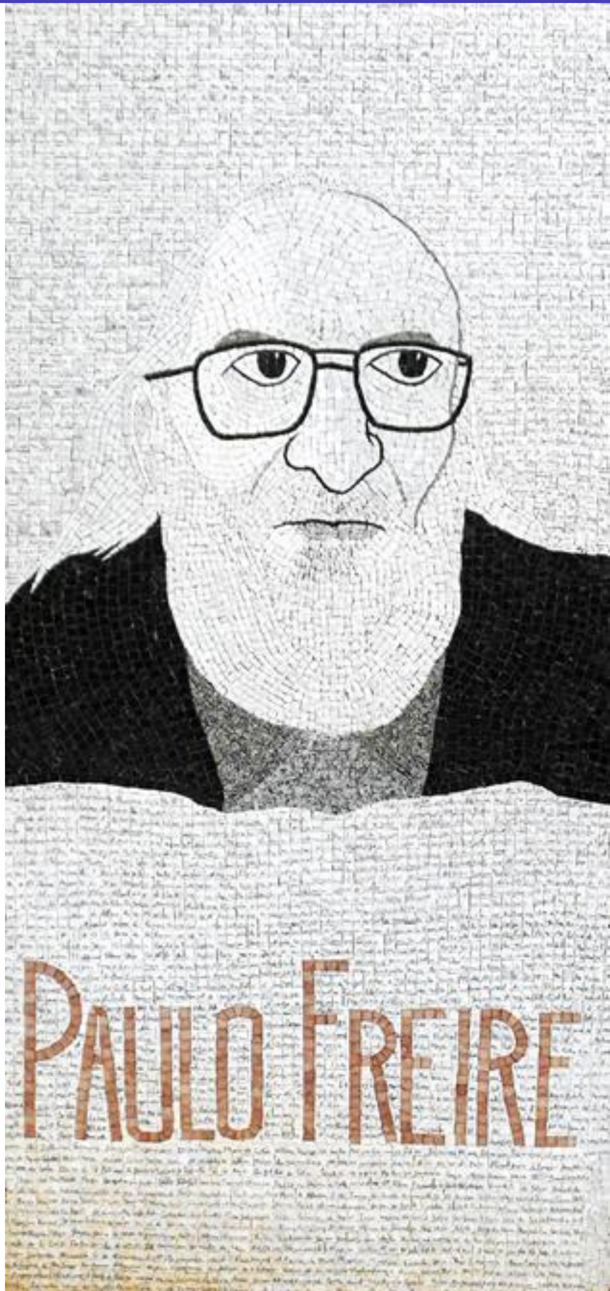
Towards clearer and more accessible science communication

As citizen participation and clear communication in science become more important, we take a look at some of REA's most interesting and informative projects in the field.

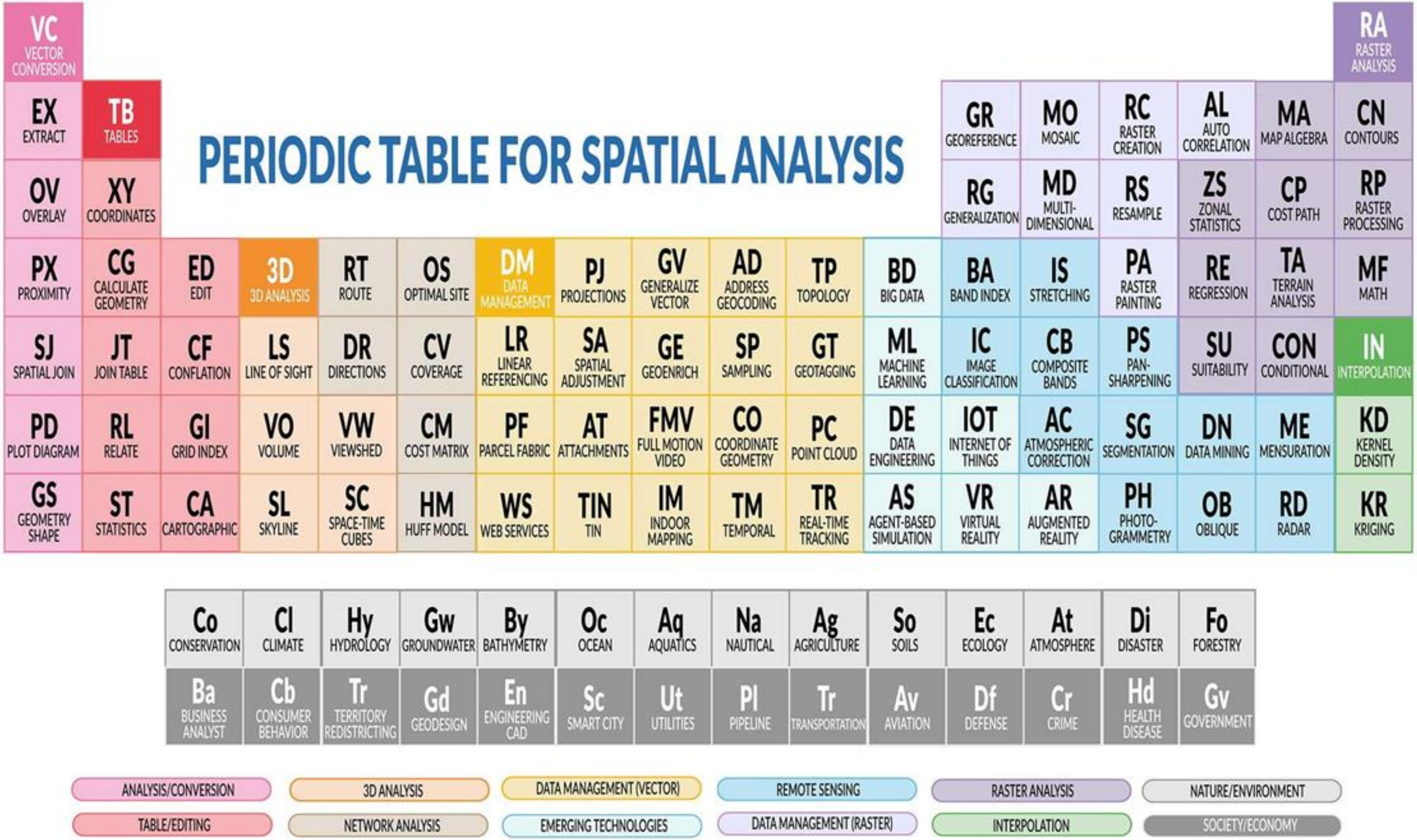
Getty Images

"Leaders who do not act dialogically, but insist on imposing their decisions, do not organize the people—they manipulate them…"

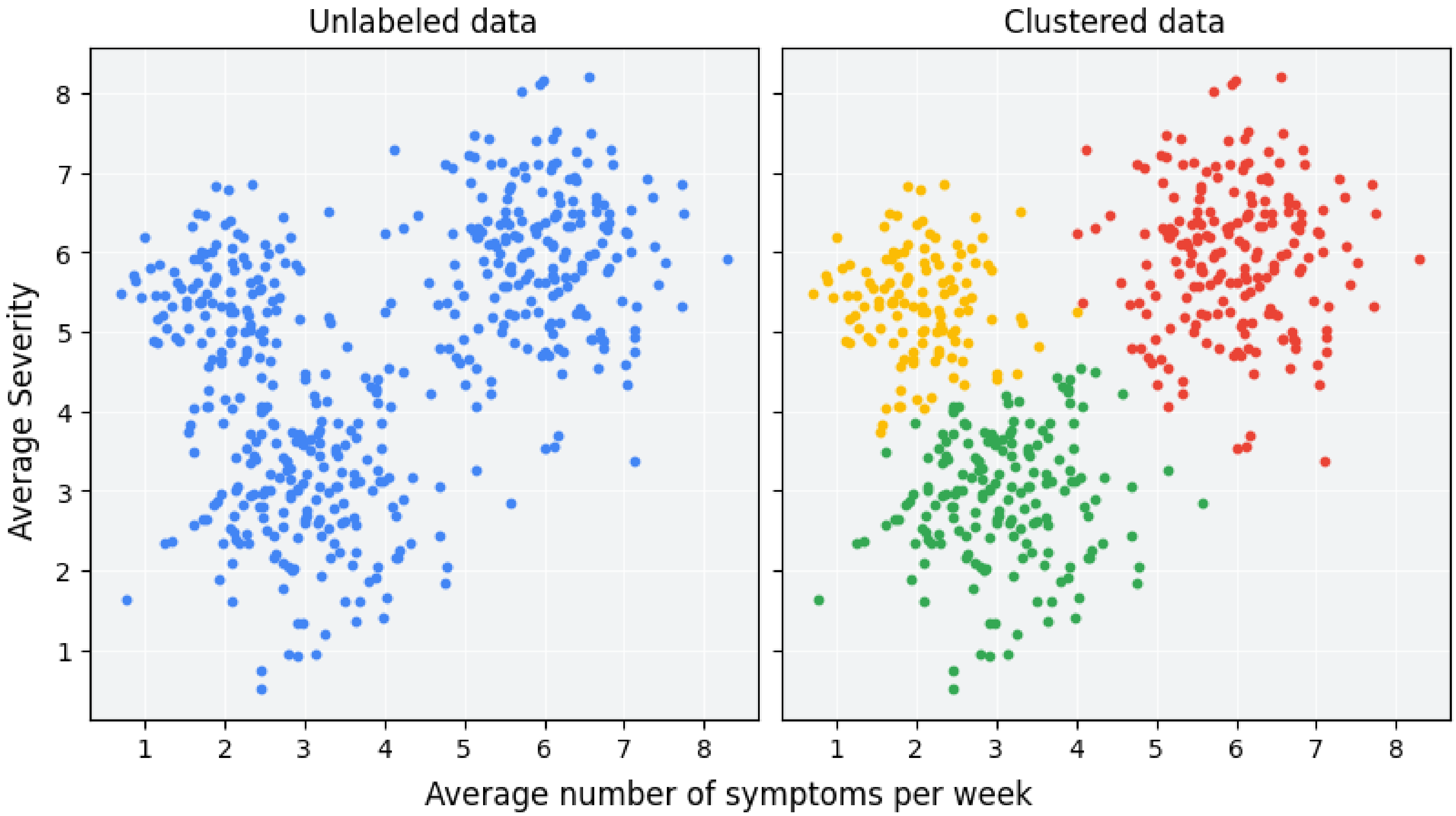
Pedagogy of the Oppressed



AVOID EXCESSIVE USE OF TECHNICAL JARGON



CLUSTERING



Source: https://developers.google.com/machine-learning/clustering/overview

Topic

Method

Joint Research Proposal -- XX minutes
PhD interests

Research Question
Roles within the team
Stakeholders
Timeline
Budget

TRANSFERRING

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Diego Pajarito Grajales | Urban Big Data Centre, University of Glasgow, UK

Co-designed with: Ana Maria Bustamante Duarte | Faculty of Architecture and Design, Universidad de Los Andes, Colombia